

SHWD-YOLO11s: Training-Policy Optimization and Cross-Domain Evaluation for Safety Helmet Detection

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Abstract—Automated safety-helmet monitoring requires both accurate localization and real-time inference under substantial domain shift. This work studies two three-head YOLO11s variants trained on the Safety Helmet Wearing Dataset (SHWD): SHWD-YOLO11s-DA, based on a data-augmentation fine-tuning policy, and SHWD-YOLO11s-LO, which modifies the box-loss weight, MixUp rate, and cosine learning-rate schedule. At 960-pixel validation resolution, the DA variant reaches 96.111% mAP50 and 63.513% mAP50–95, whereas LO reaches 96.148% and 63.760%, respectively. Measured batch-one PyTorch FP32 throughput on a Kaggle Tesla T4 is 48.7 FPS for DA and 47.0 FPS for LO. We further evaluate both checkpoints on five helmet-detection datasets using explicitly reported official, standardized, or deterministic-holdout protocols. LO improves mAP50–95 on four of five datasets, with its largest external-domain gain of 1.63 percentage points on Hard Hat Workers, but performance remains limited on SHEL5K. These results show that training-policy optimization provides modest in-domain gains and non-uniform cross-domain benefits; they do not establish a universal improvement across domains.

Index Terms—safety helmet detection, YOLO11s, data augmentation, loss optimization, cross-domain generalization, real-time inference

I. INTRODUCTION

Construction-site monitoring must identify both workers and protective helmets despite occlusion, crowding, illumination changes, and small object scale. A detector intended for operational use must therefore balance localization accuracy, recall, and inference throughput. YOLO11 provides a compact one-stage detection baseline suitable for this setting [1].

This study does not introduce an unverified re-parameterized architecture. Instead, it examines two evidence-backed training configurations applied to a checkpoint-native, three-head YOLO11s detector. The first configuration, SHWD-YOLO11s-DA, uses a hard-case augmentation fine-tuning policy. The second, SHWD-YOLO11s-LO, changes several training-policy factors: the box-loss weight increases from 7.5 to 10.0, MixUp changes from 0 to 0.15, and cosine learning-rate scheduling is enabled. Because these factors change jointly and execution of a custom Focal-EIoU implementation has not been independently verified on every distributed worker, we attribute the second configuration to policy-level loss optimization rather than to Focal-EIoU alone.

The contributions are: (1) an artifact-grounded comparison of DA and LO at 960-pixel resolution; (2) measured deployment results across GPU FP32, GPU FP16, CPU ONNX, and

a CCTV-oriented code path; and (3) a protocol-aware cross-domain benchmark covering SHWD, GDUT-HWD, SHEL5K, Hard Hat Workers, and SFCHD.

II. METHODOLOGY

A. Model Variants

Both variants retain the native YOLO11s detection topology and predict two classes, *hat* and *person*. SHWD-YOLO11s-DA uses box weight 7.5, no MixUp, and a non-cosine learning-rate schedule. SHWD-YOLO11s-LO uses box weight 10.0, MixUp probability 0.15, and cosine scheduling. Both recorded training configurations use an image size of 960×960 and device identifiers 0 and 1.

B. Evaluation Metrics

Precision P is the fraction of predicted positives that are correct, while recall R is the fraction of ground-truth objects recovered. AP is integrated over the precision–recall curve. We report mAP50 at IoU 0.5 and the stricter mAP50–95 averaged over IoU thresholds 0.50:0.05:0.95. The latter is the primary localization metric.

C. Protocol Separation

Internal validation metrics are read from the best row of each run’s `results.csv`, selected by mAP50–95. Cross-domain results use the protocol assigned to each dataset in the benchmark: official, standardized, or deterministic holdout. These protocol classes are not pooled into a single overall score because their splits and preprocessing are not necessarily exchangeable.

III. EXPERIMENTS AND RESULTS

A. Internal Validation and Runtime

Table I presents the internal results. LO raises mAP50 by 0.037 percentage points and mAP50–95 by 0.247 points relative to DA. Its precision increases by 0.773 points, while recall decreases by 0.948 points. The result therefore represents a small localization/precision improvement rather than uniform dominance.

The two variants remain above the conventional 30-FPS real-time threshold in GPU FP32. DA is 1.7 FPS faster in this path, whereas LO has the higher mAP50–95. Table II reports additional measured code paths. Because the CCTV and core-inference measurements follow different timing paths, their

TABLE I
INTERNAL SHWD VALIDATION AND MEASURED GPU FP32 THROUGHPUT. FPS IS BATCH-ONE PYTORCH FP32 AT 960×960 ON A KAGGLE TESLA T4; IT IS NOT DUAL-GPU AGGREGATE THROUGHPUT.

Model	Precision (%)	Recall (%)	mAP50 (%)	mAP50–95 (%)	Best epoch	FPS
SHWD-YOLO11s-DA	93.241	92.886	96.111	63.513	60	48.7
SHWD-YOLO11s-LO	94.014	91.938	96.148	63.760	62	47.0

TABLE II
MEASURED DEPLOYMENT THROUGHPUT (FPS) AT 960-PIXEL MODEL INPUT.

Variant	GPU FP32	GPU FP16	ONNX CPU 4T	CCTV GPU
DA	48.7	48.8	2.9	64.2
LO	47.0	47.2	3.3	70.0

latencies must not be added or interpreted as directly nested measurements.

D. Cross-Domain Generalization Benchmark

Table III reports all cross-domain outcomes. VOC2028 (SHWD) is the source-domain official validation set. GDUT-HWD Standardized and SHEL5K Standardized use benchmark-normalized formats and class mappings. Hard Hat Workers [2] uses a deterministic holdout rather than an official split. SFCHD uses the official protocol recorded by the benchmark.

On VOC2028, LO preserves mAP50 (+0.0001) and raises mAP50–95 by 0.0015; recall increases by 0.0059 while precision falls by 0.0052. On GDUT-HWD, LO improves mAP50 and mAP50–95 by 0.0067 and 0.0110, respectively, despite a 0.0098 recall reduction. SHEL5K is the weakest domain: both variants have recall near 0.375 and mAP50–95 below 0.25. LO increases SHEL5K mAP50 by 0.0024 but reduces mAP50–95 by 0.0062.

The largest external localization gain occurs on Hard Hat Workers, where LO raises mAP50–95 by 0.0163 and mAP50 by 0.0067, with a 0.0029 recall increase and a 0.0070 precision reduction. On SFCHD, LO improves recall by 0.0060, mAP50 by 0.0074, and mAP50–95 by 0.0103, while precision decreases by 0.0165. Overall, LO improves mAP50–95 on four of five datasets but does not dominate every metric or every domain. The large source-to-external decline, particularly on SHEL5K, demonstrates unresolved domain shift.

IV. THREATS TO VALIDITY

The DA and LO configurations alter multiple training factors simultaneously; their difference is not a single-factor loss ablation. No causal Focal-EIoU claim is therefore made. Cross-domain rows employ heterogeneous protocols and must not be averaged without sample-aware normalization. The deployment report provides measured values, but core-inference and CCTV paths are separate benchmark implementations. Finally, model parameters and GFLOPs are omitted until they are measured from the exact archived checkpoints.

V. CONCLUSION

SHWD-YOLO11s-DA and SHWD-YOLO11s-LO both provide real-time GPU FP32 inference at 960-pixel resolution. LO achieves the higher internal mAP50–95, with a modest 0.247-point gain over DA, and improves cross-domain mAP50–95 on four of five evaluated datasets. These improvements are accompanied by small precision–recall trade-offs and do not eliminate the pronounced domain gap on SHEL5K. Future work should verify custom-loss execution on every distributed worker, perform paired multi-seed loss ablations, and analyze localization errors by object scale and occlusion.

REFERENCES

- [1] Ultralytics, “Ultralytics yolo11 documentation,” <https://docs.ultralytics.com/models/yolo11/>, accessed 13 September 2026.
- [2] A. MVD, “Hard hat workers dataset,” Public dataset release, evaluated using a deterministic holdout in this study.

TABLE III
CROSS-DOMAIN GENERALIZATION BENCHMARK. DA DENOTES DATA AUGMENTATION AND LO DENOTES LOSS OPTIMIZE. PROTOCOLS ARE REPORTED RATHER THAN POOLED.

Model	Dataset	Protocol	Prec.	Rec.	mAP50	mAP50–95
SHWD-YOLO11s-DA	VOC2028 (SHWD)	Official	0.9213	0.8871	0.9332	0.7229
	GDUT-HWD Standardized	Standardized	0.9418	0.6775	0.7359	0.3801
	SHEL5K Standardized	Standardized	0.9315	0.3755	0.4072	0.2498
	Hard Hat Workers (AndrewMVD)	Deterministic holdout	0.9393	0.6870	0.7373	0.4222
	SFCHD	Official	0.8130	0.5515	0.6406	0.3632
SHWD-YOLO11s-LO	VOC2028 (SHWD)	Official	0.9161	0.8930	0.9333	0.7244
	GDUT-HWD Standardized	Standardized	0.9454	0.6677	0.7426	0.3911
	SHEL5K Standardized	Standardized	0.9245	0.3743	0.4096	0.2436
	Hard Hat Workers (AndrewMVD)	Deterministic holdout	0.9323	0.6899	0.7440	0.4385
	SFCHD	Official	0.7965	0.5575	0.6480	0.3735